INCH-POUND

MIL-M-12699B(AR)
10 September 1990
SUPERSEDING
MIL-M-12699A(Ord)
02 November 1954

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M25 AND M25E1

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mount used with telescope, panoramic, M12 for laying various artillery weapons in elevation for direct or indirect fire.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specifications to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	-Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	-Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg.4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

DRAWINGS

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

6523553	-Fixture, Elevation & Cross level Testing
7295505	-Perpendicularity Gage
7570429	-Mount, Telescope: M25
8213353	-Mount, Telescope: M25E1
8658940	-Gunner Quadrant, Calibrated

PACKAGING DATA SHEET

7578429	-Packaging of Mount, Telescope: M25
8213353	-Packaging of Mount, Telescope: M25E1

(Copies of drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.1).

3.2 Materiels. Materiels shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

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3.3 Fabrication. The M25 mount shall be manufactured in accordance with Drawing 7578429 and drawings pertaining thereto. The M25E1 mount shall be manufactured in accordance with Drawing 8213353 and drawings pertaining thereto.

3.4 General specifications. The contractor shall be responsible for the compliance with the requirements of specification MIL-F-13926.

3.5 Vibration. The mount shall be vibrated between 2 to 2-1/2 minutes at an amplitude of 1/16 inch (1/8 inch total movement) and a frequency of 30 cycles per second after which it shall meet all requirements.

3.6 Adjustment. With the mount set up on a leveled test fixture, and with the plane of the mounting pads of the bracket perpendicular to the axis of rotation of the fixture, the actuating arm bracket clamped so that the leveling pads of the arm bracket are horizontal and the axis of the telescope socket vertical, the mount shall be adjusted as follows:

3.6.1 Axis of telescope socket. The axis of telescope socket shall be vertical within (1 mil) 3 minutes, 22.5 seconds, and top surface of telescope socket shall be horizontal within one minute of arc.

3.6,2 Elevation level vial bubble. The elevation level vial bubble shall be centered.

3.6.3 Elevation scale and micrometer. The elevation scale and micrometer shall be adjusted so that the zero graduations coincide with their respective indexes.

3.6.4 Line of sight. The adjusting screws in the telescope socket shall be set to bring the line of sight of a properly adjusted panoramic telescope perpendicular to the center line of the mounting bore within plus or minus 0.1 mil.

3.6.5 Cross leveling of mount. The mount shall be cross leveled to the position where the deviation of the line of sight from a plumb line target is within plus or minus 0.2 mil when the fixture is rotated from zero to 1100 mils elevation for the M25 Mount, and zero to 1300 mils for the M25E1 Mount, with the cross' level vial bubble centered.

3.7 Accuracy. With the mount adjusted as specified in 3.6, the following conditions shall be met:

3.7.1 Elevation accuracy. The elevation mechanism error shall not exceed 1.0 mil excluding backlash.

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3.7.2 Plumb travel. With the line of sight in coincidence with a plumb line, the line of alight shall follow the plumb line within plus or minus 0.75 mil (total spread 1.5 mils) when the M25 mount is rotated from zero to 1100 mils for the M25 mount, and zero to 1300 mils for the M25E1 mount, by means fo teh elevating worm.

3.7.3 Elevation backlash. The backlash in the elevation worm mechanism shall not exceed 0.5 mil.

3.7.4 Azimuth correction. The mount shall give azimuth correction readings in accordance with the table given below after the following conditions are met in sequence as listed:

(a) Angle of cant (cross level) set to seven degrees

(b) Elevation angle measured in a vertical plane as indicated on telescope mount scale and micrometer

(c) Cross level and horizontal level bubbles centered by means of the cross level knob and knob actuating elevation setting of test fixture

Angle of Cant (Cross Level)	Elevation Angle Measured On Mount, Elev. Scale	Azimuth Correction Reading	Tolerance Plus or Minus
Degrees	Mils	Mils	Mils
7	500	66.9	1.0
7	800	125.4	1.5
7	1100	236.1	2.0

3.8 Torque. The maximum torque required to rotate the following knobs shall not exceed the values specified as follows:

(a) Elevation mechanism knob - 14 pound-inches

(b) Cross leveling knob - 12 pound-inches

3.9 Operating temperatures. The mount shall be operable at -40° ± 5° Fahrenheit (F) (-40° ± 2° Centigrade (C)) and +150° ± 5° F (+65° ± 2° C). Subsequent to the above temperature extremes the mount shall meet all the requirements of this specification.

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3.10 Workmanship. The workmanship of MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence, of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial Production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing Shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by Quality Assurance Provisions (QAPs). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The three (3) mounts shall be tested in accordance with, and meet the requirements of, Tables I and II.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this

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inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Examinations and tests.

4.3.1 Components and subassemblies. All components and subassemblies shall be inspected an accordance with the inspection provisions contained in QAPs listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.3 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

4.3.4 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor’s proposed alternative(s) provide equal assurance, the provision of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor’s quality program or detailed inspection system, as applicable.

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TABLE I. Requirements and test procedures.

No.	Characteristic	Requirement	Test procedu
Critical:	None		
Major:	100% Inspection		
101.	Vibration	3.5	4.6.1
102.	Axis of telescope socket	3.6.1	4.6.2
103.	Elevation level vial bubble	3.6.2	4.6.3
104.	Elevation scale and micrometer	3.6.3	4.6.4
105.	Line of sight	3.6.4	4.6.5
106.	Cross leveling of mount	3.6.5	4.6.6
107.	Elevation accuracy	3.7.1	4.6.7
108.	Plumb travel	3.7.2	4.6.8
109.	Elevation backlash	3.7.3	4.6.9
110.	Azimuth correction	3.7.4	4.6.10
111.	Torque	3.8	4.6.11
112.	Workmanship	3.10	MIL-F-13926
113.	Packaging	5.1	4.7
Minor:	None		

4.4 Special sampling.

4.4.1 General. Subsequent to meeting the requirements of Table I, three quadrants shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table I after being subjected to Table II testing.

TABLE II. Requirements and test procedures

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Operating temperature	3.9	

4.4.2 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Test equipment.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise stated in the contract, all inspection equipment specified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the list of Inspection Documents when provided with the TDP.

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the requirements of MIL-F-13926. Since the tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment (see 6.3).

4.5.3.3 Test equipment. In conjunction with 4.5.3.2, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.11.

NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracy outlined under “Test facilities” in MIL-F-13926.
2. Torque wrench	Standard shop type accurate to 10% of drawing tolerance.

4.6 Methods of inspection.

4.6.1 Vibration. The mount shall be vibrated for the duration and at the frequency and amplitude specified in 3.5.

4.6.2 Axis of telescope socket. The axis of the socket shall be tested by utilizing Perpendicularity Gage, 7295505, except that find number 1, Body, shall be 5 1/2 inches in diameter and the top surface shall be held flat and parallel to within .0002 of the 9/32 dimension surface and the top surface to be

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perpendicular to the .7495 diameter, Datum “A” to within .0002. (The "V” does not have to be machined into find 1.) The gage shall be inserted in the socket and calibrated gunners quadrant, 8658940, shall be used to verify that the axis of the socket meets the requirement of 3.6.1, there shall be 2 readings taken 900 apart. The gunners quadrant shall be placed on a six in parallel and two readings, 90° apart shall be taken on the top of the socket and shall meet the requirements of 3.6.1.

4.6.3 Elevation level vial bubble. The elevation level vial bubble shall be centered to meet the requirements of 3.6.2.

4.6.4 Elevation scale and micrometer. The elevation scale shall be lined up to its respective lndice and the micrometer scale shall be lined up to its respective indite to meet the requirements of 3.6.3.

4.6.5 Line of sight. The line of sight shall be adjusted with the adjusting screws to meet the requirements of 3.6.4.

4.6.6 Cross leveling of mount. The mount shall be cross leveled in accordance with 3.6.5 and the cross level bubble shall remain centered.

4.6.7 Elevation accuracy. The elevation accuracy shall be tested at zero mils and at 200 mil increments, and at 1100 mils for the M25 mount, and 1300 mils for the M25E1 mount, and shall meet the requirements of 3.7.1.

4.6.8 Plumb travel. The travel shall be tested in accordance with, and shall meet the requirements of 3.7.2.

4.6.9 Elevation backlash. Place the modified gage, 7295505, in the socket of the mount. Turn the elevation worm gear in a riqht hand direction until the zero on the micrometer scale lines up with indite on the body at zero elevation; do not go past zero. Take ii reading with the gunner’s quadrant, 8658940, and record this reading. Turn the knob one full turn in the right hand direction. Reverse the direction of the knob until the zero lines up with the indite, do not go past zero. Take a reading with the gunner's quadrant. The difference between the two readings is the backlash. Perform this test at the increments of 4.6.7, and shall meet the requirements of 3.7.3.

4.6.10 Azimuth correction. The mount level vial bubbles shall be centered after cant and elevation settings have been established, the azimuth correction readings taken from the telescope and shall be within the tolerances of 3.7.4.

4.6.11 Torque. The elevation and cross leveling knobs shall be turned a minimum of 3 complete revolutions in a clockwise

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direction, and 3 complete revolutions in a counterclockwise direction, and shall not exceed pound-inches of 3.8.

4.7 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of 5.1.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 7578429 for the M25 mount and 8213353 for the M25E1 mount. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Procurement data should specify the following:

a. Title, number and date of this specification.

b. Selection of an applicable level of preservation, packaging, and packing (See 5.1).

c. Applicable packaging data sheet number (See 5.1).

d. Applicable stock number.

e. Provisions for First Article Testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-1714. (See 6.2.1.)

6.2. Inspection equipment design.

6.2.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the data of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the

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contract. Unless otherwise specified, data item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.4 Subject term (key word) listing.

Fire control
Mount

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-A959)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-12699B (AR)	10 September 1990

3. DOCUMENT TITLE
MOUNT, TELESCOPE: M25 AND M25E1

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

	DATE SUBMITTED

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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